15.075 Exam 1

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Grading is based on demonstration of conceptual understanding, so you need to show all of your work.

Problem 1

A very large bin contains 3 different types of disposable flashlights. The probability that a type 1 flashlight will give over 100 hours of use is 0.7, with the corresponding probabilities for type 2 and 3 flashlights being 0.4 and 0.3 respectively. Suppose that 20% of the flashlights in the bin are type 1, 30% are type 2 and 50% are type 3.

1. What is the probability that a randomly chosen flashlight will give more than 100 hours of use?
2. Given the flashlight lasted over 100 hours, what is the conditional probability that it was a type 1 flashlight?

Solution

a. Let

Problem 3

The annual snowfall at a particular city is modeled as a normal random variable with a mean of $\mu = 60$ inches and a standard deviation of $\sigma = 20$. What is the probability that this year's snowfall will be between 40 and 100 inches?

Solution

Let X the annual snowfall. It is:

$$P(40 \leq X \leq 100) = P(\frac{40-60}{20} \leq \frac{X-60}{20} \leq \frac{100-60}{20}) = P(-1 \leq Z \leq 2) = \Phi(2) - \Phi(-1) = 0.8186.$$

Problem 4

An airline company accepts 100 reservations for a flight that has 72 available seats. A person booking a reservation does not show up in the flight with probability $q = 0.1$. Find an upper bound in the probability that no traveler showing up is bumped (excluded) from the flight.

Solution

Let X be the number of people showing up in the flight. Then X is a $Bin(n, p)$ with $n = 100$ and $p = 0.9$. It is $E[X] = n \cdot p = 100 \cdot 0.9 = 90$ and $var(X) = np(1-p) = 9$. We are interested in $P(X \leq 72)$. We have:

$$P(|X - 90| \geq 18) = P(X \geq 108) + P(X \leq 72) = P(X \leq 72)$$

But from Chebyshev's inequality we know:

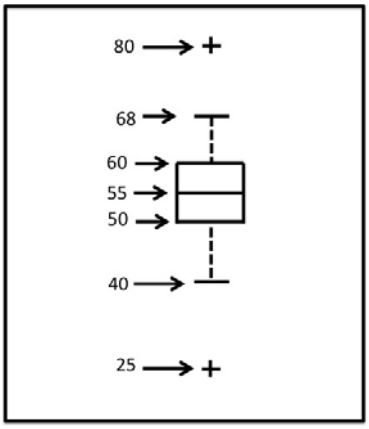
$$P(|X - E[X]| \geq 18) \leq \frac{var(X)}{18^2} = \frac{1}{36}.$$

Therefore,

$$P(X \leq 72) \leq \frac{1}{36}.$$

Problem 5

Provide a sample that could correspond to this box plot:



Solution

Obviously, there are uncountably infinite solutions to this problem. Two representative samples are: $\{25, 40, 50, 52, 53, 55, 56, 58, 60, 68, 80\}$ and $\{25, 40, 48, 50, 51, 52, 53, 55, 56, 57, 58, 60, 65, 68, 80\}$

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